



## LAB Task :1

| Only for course Teacher     |                   |            |            |               |            |
|-----------------------------|-------------------|------------|------------|---------------|------------|
|                             | Needs Improvement | Developing | Sufficient | Above Average | Total Mark |
| Allocate mark & Percentage  | 25%               | 50%        | 75%        | 100%          | 20         |
| Clarity                     |                   |            |            |               |            |
| Content Quality             |                   |            |            |               |            |
| Spelling & Grammar          |                   |            |            |               |            |
| Organization and Formatting |                   |            |            |               |            |
| Total obtained mark         |                   |            |            |               |            |
| Comments                    |                   |            |            |               |            |

**Semester: Fall 2026**

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**Batch: 46**

**Section: B**

**Course Code: CSE 124**

**Course Name: Object Oriented Programming Lab Language Lab**

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**Designation: Lecturer**

**Submission Date: 30/ 07/ 2026**

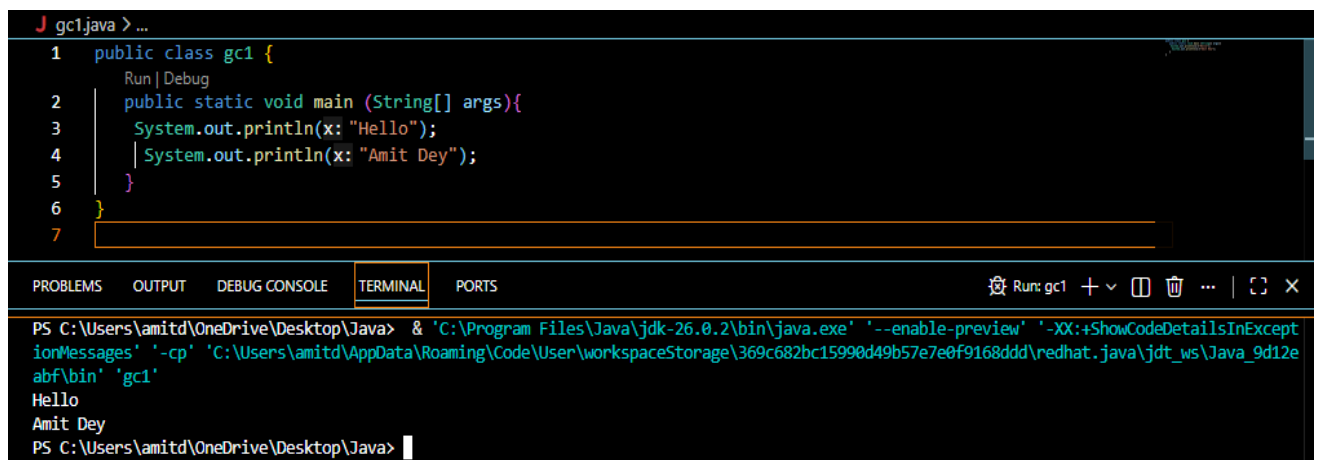
**1. Write a Java program to print 'Hello' on screen and your name on a separate line.**

:

Hello

Amit Dey

**ANSWER:**



The screenshot shows an IDE with a Java class named `gc1` containing a `main` method. The code prints "Hello" and "Amit Dey" on separate lines. The terminal window below shows the command to run the program, which successfully executes and displays the expected output.

```
J gc1.java > ...  
1 public class gc1 {  
    Run | Debug  
2     public static void main (String[] args){  
3         System.out.println(x: "Hello");  
4         System.out.println(x: "Amit Dey");  
5     }  
6 }  
7  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
PS C:\Users\amitd\OneDrive\Desktop\Java> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\amitd\AppData\Roaming\Code\User\workspaceStorage\369c682bc15990d49b57e7e0f9168ddd\redhat.java\jdt_ws\Java_9d12eabf\bin' 'gc1'  
Hello  
Amit Dey  
PS C:\Users\amitd\OneDrive\Desktop\Java>
```

**Explain:**

I learned how to declare a Java class and the `main()` method. I also learned how to use `System.out.println()` to print text on the console.

## 2. Sum of Two Numbers

Write a Java program to print the sum of two numbers.

Test Data:

 $74 + 36$ 
$$\vdots$$

110

**ANSWER:**

```
J gc2.java > gc2 > main(String[])

1  class gc2{
    Run | Debug
2  |  public static void main(String[] args){
3  |      int a = 74;
4  |      int b = 36;
5  |      int sum = a+b;
6  |      System.out.println(sum);
7  |  }
8  |  }
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

PS C:\Users\amitd\OneDrive\Desktop\Java> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\amitd\AppData\Roaming\Code\User\workspaceStorage\369c682bc15990d49b57e7e0f9168ddd\redhat.java\jdt\_ws\Java\_9d12eabf\bin' 'gc2'

110

PS C:\Users\amitd\OneDrive\Desktop\Java> |

**Explain:**

I learned how to declare integer variables and store values in them. And I also learned how to add two numbers using the `+` operator and display the result with `System.out.println()`.

### 3. Division of Two Numbers

Write a Java program to divide two numbers and print them on the screen.

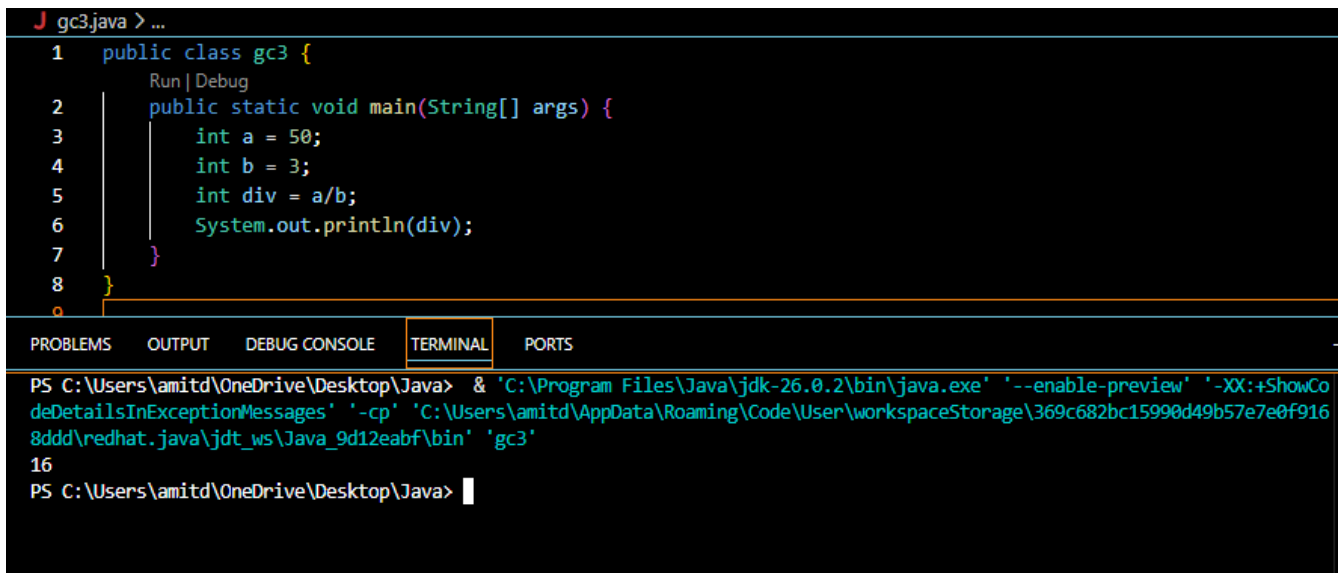
Test Data:

50/3

:

16

#### ANSWER:



```
J gc3.java > ...
1 public class gc3 {
    Run | Debug
2     public static void main(String[] args) {
3         int a = 50;
4         int b = 3;
5         int div = a/b;
6         System.out.println(div);
7     }
8 }
9

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\amitd\OneDrive\Desktop\Java> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\amitd\AppData\Roaming\Code\User\workspaceStorage\369c682bc15990d49b57e7e0f9168ddd\redhat.java\jdt_ws\Java_9d12eabf\bin' 'gc3'
16
PS C:\Users\amitd\OneDrive\Desktop\Java> |
```

#### Explain:

I learned how to perform division using the `/` operator in Java. I learned how to store the result in a variable and print it using `System.out.println()`.

## 4. Arithmetic Operations

Write a Java program to print the results of the following operations.

a.  $-5 + 8 * 6$

b.  $(55+9) \% 9$

c.  $20 + -3*5 / 8$

d.  $5 + 15 / 3 * 2 - 8 \% 3$

:

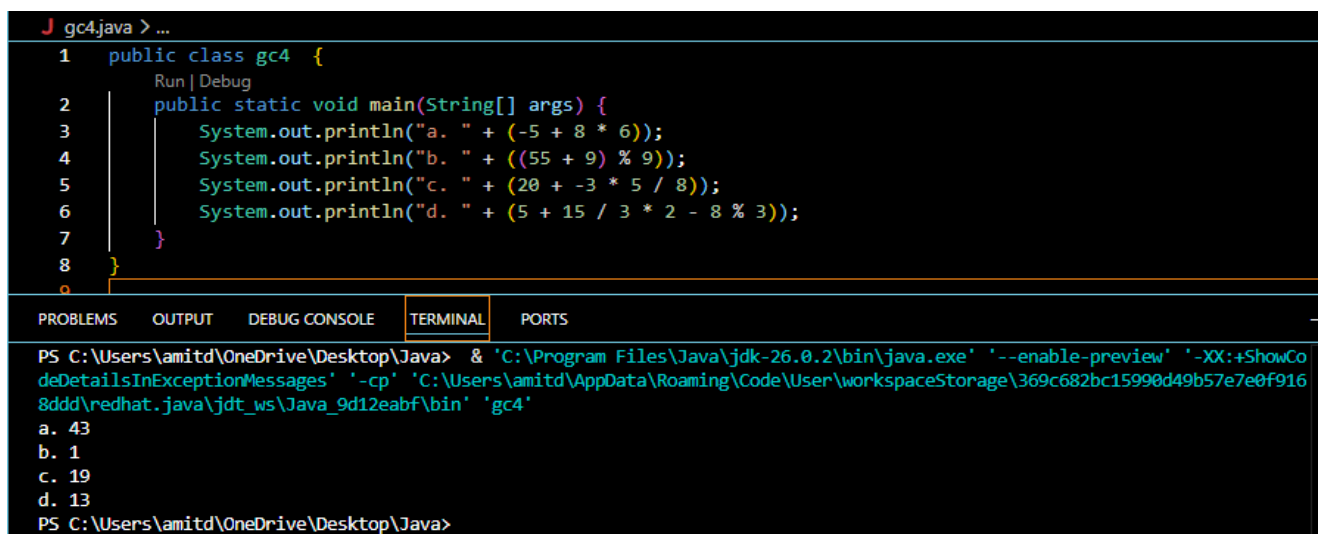
43

1

19

13

### ANSWER:



The screenshot shows an IDE with a Java class named `gc4`. The code defines a `main` method that prints the results of four arithmetic operations. The output window shows the results: a. 43, b. 1, c. 19, and d. 13.

```
1 public class gc4 {  
2     public static void main(String[] args) {  
3         System.out.println("a. " + (-5 + 8 * 6));  
4         System.out.println("b. " + ((55 + 9) % 9));  
5         System.out.println("c. " + (20 + -3 * 5 / 8));  
6         System.out.println("d. " + (5 + 15 / 3 * 2 - 8 % 3));  
7     }  
8 }
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

```
PS C:\Users\amitd\OneDrive\Desktop\Java> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCo  
deDetailsInExceptionMessages' '-cp' 'C:\Users\amitd\AppData\Roaming\Code\User\workspaceStorage\369c682bc15990d49b57e7e0f916  
8ddd\redhat.java\jdt_ws\Java_9d12eabf\bin' 'gc4'  
a. 43  
b. 1  
c. 19  
d. 13  
PS C:\Users\amitd\OneDrive\Desktop\Java>
```

### Explain:

I learned how to evaluate arithmetic expressions using operators such as  $+$ ,  $-$ ,  $*$ ,  $/$ , and  $\%$ . I learned that Java follows **operator precedence** and evaluates expressions accordingly.

## 5. Product of Two Numbers

Write a Java program that takes two numbers as input and displays the product of two numbers.

Input first number: 25

Input second number: 5

$$\vdots$$
$$25 \times 5 = 125$$

**ANSWER:**

```
J gc5.java > ...
1 public class gc5 {
    Run | Debug
2     public static void main(String[] args) {
3         int first_number = 25;
4         int second_number = 5;
5         System.out.println(first_number + " x " + second_number + " = " + (first_number * second_number));
6     }
7 }
8
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

PS C:\Users\amitd\OneDrive\Desktop\Java> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\amitd\AppData\Roaming\Code\User\workspaceStorage\369c682bc15990d49b57e0f9168ddd\redhat.java\jdt\_ws\Java\_9d12eabf\bin\' 'gc5'

25 x 5 = 125

PS C:\Users\amitd\OneDrive\Desktop\Java>

**Explain:**

I learned how to declare variables and store numeric values in Java. I learned how to multiply two numbers using the `*` operator and print the result in a formatted output using `System.out.println()`.

## 6. Basic Arithmetic Operations

Write a Java program to print the sum (addition), multiply, subtract, divide and remainder of two numbers.

Input first number: 125

Input second number: 24

$$\vdots$$
$$125 + 24 = 149$$
$$125 - 24 = 101$$
$$125 \times 24 = 3000$$
$$125 / 24 = 5$$
$$125 \bmod 24 = 5$$

**ANSWER:**

The screenshot displays an Integrated Development Environment (IDE) window. The top pane shows a Java file named `J gc6.java > ...`. The code defines a public class `gc6` with a `main` method. Inside `main`, several integer variables are declared and assigned values: `first_number = 125`, `second_number = 24`, `sum = first_number + second_number`, `subtract = first_number - second_number`, `multiplication = first_number * second_number`, `division = first_number / second_number`, and `modulus = first_number % second_number`. Each calculation is followed by a `System.out.println` statement to print the result.

The bottom pane is divided into two sections. On the left, there's a tabbed interface with tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL (which is active), and PORTS. Below these tabs, the terminal shows the command prompt running the Java program:

```
PS C:\Users\amitd\OneDrive\Desktop\Java> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\amitd\AppData\Roaming\Code\User\workspaceStorage\369c682bc15990d49b57e70f9168ddd\redhat.java_jdt_ws\Java_9d12eabf\bin\' 'gc6'
```

On the right side of the terminal, there's a status bar indicating "Run: gc6" along with icons for adding/removing configurations, opening/closing windows, and other standard IDE controls.

**Explain:**

I learned how to store the results of arithmetic operations (+, -, \*, /, %) in separate variables. I learned how to display multiple calculated results using `System.out.println()`.

## 10. Formula Computation

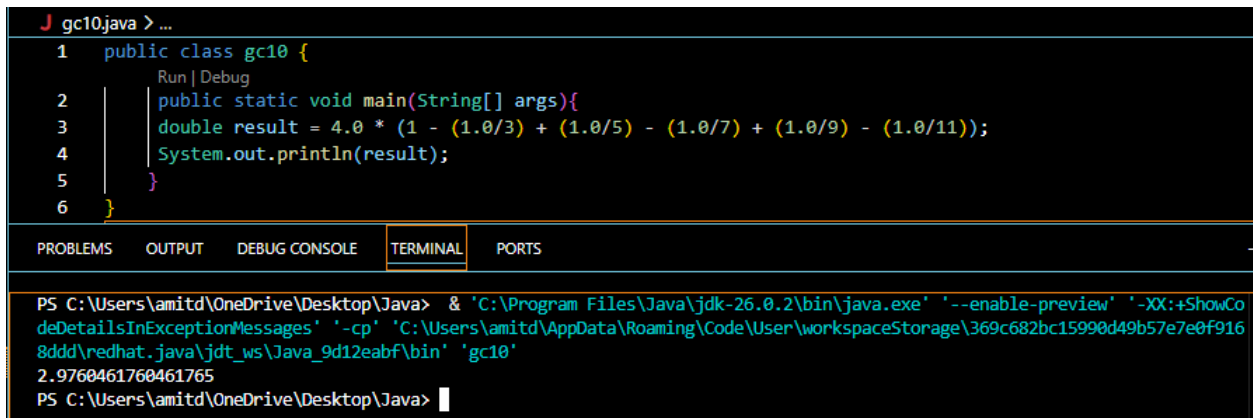
Write a Java program to compute a specified formula.

$4.0 * (1 - (1.0/3) + (1.0/5) - (1.0/7) + (1.0/9) - (1.0/11))$

:

2.9760461760461765

### ANSWER:



The screenshot shows an IDE with a Java file named `gc10.java`. The code defines a public class `gc10` with a `main` method. Inside the `main` method, a `double` variable `result` is calculated using the formula  $4.0 * (1 - (1.0/3) + (1.0/5) - (1.0/7) + (1.0/9) - (1.0/11))$ , and the result is printed to the console. The IDE interface includes tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The TERMINAL tab is active, showing the command to run the program and the output `2.9760461760461765`.

```
J gc10.java > ...
1 public class gc10 {
    Run | Debug
2     public static void main(String[] args){
3         double result = 4.0 * (1 - (1.0/3) + (1.0/5) - (1.0/7) + (1.0/9) - (1.0/11));
4         System.out.println(result);
5     }
6 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\amitd\OneDrive\Desktop\Java> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\amitd\AppData\Roaming\Code\User\workspaceStorage\369c682bc15990d49b57e7e0f9168ddd\redhat.java\jdt_ws\Java_9d12eabf\bin' 'gc10'
2.9760461760461765
PS C:\Users\amitd\OneDrive\Desktop\Java>
```

### Explain:

I learned how to use `double` for decimal calculations in Java. I learned how to evaluate complex mathematical expressions using arithmetic operators and parentheses.